

Quantifying the Spatial Reach of Perturbation Effects in Tumor CRISPR Screens

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Abstract

Spatial CRISPR screens connect genetic perturbations with phenotypes measured in intact tissue, but a molecular gradient around guide-positive cells can reflect source geometry, preferential niche occupancy or a genuine non-cell-autonomous response. Here we developed a clone-centric analysis that separates reconstruction of the perturbation source field from measurement of surrounding transcriptional states. We applied the framework to Rab8a perturbations in six SPAC-seq tumour sections. Malignant guide-positive bins defined the source, a fixed 16- μ m Gaussian kernel reconstructed only the guide field, and target-cell expression remained unsmoothed. Thirty-three Rab8a source components were identified across one lung-metastasis and five subcutaneous sections. Hypoxia associations were positive in all six sections and Rab8a-associated neighbourhoods were generally endothelial-low. The mean standardized Rab8a-hypoxia association was 0.0175, whereas the median of a structured spatial-label null was 0.0133 (permutation $P = 0.170$, false-discovery rate $q = 0.647$). Thus, Rab8a-labelled territories reproducibly occupy a hypoxia-high spatial niche, but current evidence does not establish a causal non-cell-autonomous response or a propagation radius. These findings define an experimentally testable Rab8a spatial phenotype and illustrate why spatial source geometry and structured null models are essential for interpreting in situ CRISPR screens.

1 Introduction

Spatial transcriptomics has made it possible to measure molecular state while retaining tissue architecture[1, 2]. Pooled CRISPR screens with transcriptomic or imaging readouts have, in parallel, enabled genetic perturbations to be resolved at single-cell scale[3–5]. Combining these principles enables perturbation identities to be related to nearby immune, stromal and malignant-cell states in native tumour tissue[6, 7]. The central inferential challenge is that several distinct processes can produce similar spatial maps. A perturbation clone has a physical extent; edited cells may preferentially occupy a pre-existing niche; neighbouring cells may show a local state associated with the source; and a perturbation-dependent signal may propagate and decay with distance. Only the final process naturally defines a biological propagation radius.

Spatial neighbourhood methods already demonstrate that gene expression can be associated with kernel-defined tissue context without establishing causal direction[8]. The same problem is amplified in spatial CRISPR data, where guide dropout may fragment one territory into several apparent components and thousands of neighbouring bins may create pseudoreplication[9]. An interpretable analysis must therefore define source territories explicitly, preserve the spatial structure of the null, and use the clone, section and animal—rather than the individual bin—as the relevant replication hierarchy.

Here we use Rab8a as a focused case study. We reconstruct malignant guide-positive source territories, smooth only the sgRNA source field, retain target expression in its measured form, and quantify response programmes as a function of distance from the source. We then compare the observed association with a structured spatial-label null. This de-

sign tests whether Rab8a-labelled territories occupy reproducible molecular neighbourhoods while avoiding an unsupported causal radius claim.

2 Results

2.1 A context-adjusted framework separates source and spillover associations

The framework integrates gene expression, cell-type labels, perturbation identities and spatial coordinates (Fig. 1). Expression and cell-type neighbourhoods are encoded separately through pooling and multilayer perceptrons and then combined into a context representation. This representation defines a matched-control pool within the same section, cell type and niche while excluding spatially proximal or guide-positive candidates. The matched controls are intended to approximate the local background state against which source-associated and neighbourhood-associated responses are compared.

Spatial association is represented with two complementary variables. A source indicator identifies bins belonging to a perturbation territory, whereas an additive exposure field accumulates distance-weighted contributions from all source clones. A context-adjusted regression estimates the source association and the surrounding spillover association jointly, conditional on the encoded context. This separation prevents the molecular state of a guide-positive source from being conflated with the response of neighbouring non-source bins.

For the Rab8a case study, guide assignment was restricted to bins with at least two guide UMIs and a top-guide fraction of at least 0.70. Source candidates were further restricted to

malignant-like bins, after which spatial components were identified within each section and target gene. A fixed 16- μm Gaussian kernel was used to reconstruct the source guide field. Importantly, this kernel was not applied to gene expression or pathway scores.

Target bins are assigned to their nearest Rab8a source component, preventing the same bin from being counted repeatedly for multiple Rab8a components. Response programmes are summarized in 0–20, 20–40, 40–80, 80–120 and 120–200- μm rings. The final ring is used only as a within-section far-field comparator. It is not interpreted as evidence for a 200- μm biological effect.

The analysis included six tumour sections: M001 from a lung-metastasis cohort and subQ-1 to subQ-5 from a subcutaneous cohort. Thirty-three Rab8a source components passed the working filters. All carried the same detected guide, sgRab8a_1, precluding independent-guide concordance analysis.

This source-first design was chosen to keep four quantities conceptually separate. The observed guide-positive bins describe directly detected perturbation signal; the KDE core describes the reconstructed spatial support of that signal; the distance rings describe the geometry of surrounding tissue; and the response programme describes the molecular state of target bins. Consequently, an apparent response gradient cannot arise merely because the target expression values were smoothed with the same kernel used to call the source. The framework also records the component and section identities of every target-bin assignment, allowing the statistical summaries to be calculated above the individual-bin level.

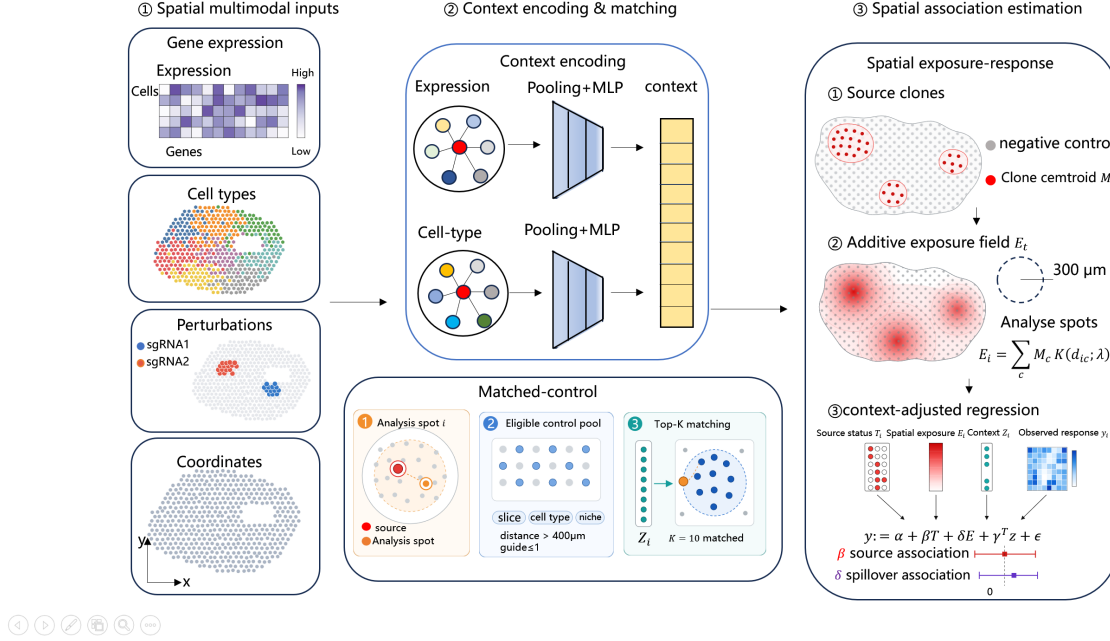


Figure 1. Context-adjusted framework for estimating source and spatial spillover associations. **Step 1, spatial multimodal inputs.** The model receives gene-expression profiles, cell-type assignments, perturbation identities and physical coordinates for the analysed spatial bins. **Step 2, context encoding and matching.** Expression and cell-type neighbourhoods are encoded through modality-specific pooling and multilayer perceptrons and combined into a context vector. For each analysis bin, eligible controls are restricted to the same section, cell type and niche; candidates within 400 μm or carrying more than one guide assignment are excluded. The ten nearest candidates in context space form the matched-control set. **Step 3, spatial association estimation.** Guide-positive source bins are grouped into source clones. An additive exposure field, $E_i = \sum_c M_c K(d_{ic}; \lambda)$, accumulates the distance-weighted contribution of clone c at analysis bin i ; bins within 300 μm of a source are included in the local exposure analysis. Context-adjusted regression, $y_i = \alpha + \beta T_i + \delta E_i + \gamma^T z_i + \epsilon_i$, separates the source association β from the spatial spillover association δ , conditional on context z_i . The diagram defines the general model architecture; causal interpretation additionally requires independent guides, matched controls and biological replication.

2.2 Rab8a territories repeatedly occupy hypoxia-high tumour regions

Spatial maps showed that Rab8a source bins and KDE-defined source cores were embedded in regions with elevated raw hypoxia scores (Fig. 2a). The same colour scale, capped at the 99th percentile, was used across sections to prevent a small number of extreme values from dominating the display. Source coordinates, source-field contours and centroids remained visible on the tissue coordinate system, allowing the association to be inspected independently of the model summary.

All six section-level Rab8a–hypoxia coefficients were positive. The mean standardized component association was 0.0175, and a descriptive section bootstrap interval ranged from approximately 0.0098 to 0.0271. The same-sign result across six sections corresponds to a descriptive exact sign probability of 0.03125 under an independent fair-sign model. Because section-to-animal mapping was unavailable, this value is not treated as a biological-replicate inferential P value.

Rab8a-associated neighbourhoods were also generally endothelial-low. The joint spatial coefficient for the endothelial programme was -0.0151 . Together, the two directions describe a hypoxia-high, endothelial-low spatial state rather than a single isolated pathway change.

The spatial overview further showed that the association was not confined to a single visually selected component. Rab8a-positive bins were distributed across sections with different tissue outlines and local densities, yet high-hypoxia neighbourhoods recurred around source fields in both cohorts. Representative component maps preserved the measured source points and displayed the reconstructed KDE boundary separately (Fig. 2b). This distinction

was important in regions with sparse or discontinuous guide detection, where an interpolated source field improved geometric continuity without changing the underlying target measurements.

The radial profiles provided a second representation of the same observation (Fig. 3a). In M001, for example, the mean raw hypoxia score was 0.600 in the 0–20- μm ring and 0.258 in the 120–200- μm far field, whereas the corresponding endothelial scores were 0.046 and 0.191. These values illustrate the direction of the local contrast in one section; they are not treated as independent tests because bins within a section share tissue context and source geometry. Across components, the number of eligible target bins varied with tissue coverage and component position, and this support was retained in the source-data tables rather than replacing missing rings with zero.

The complete component audit confirmed substantial variation in source size, local tissue support and radial curve shape across the 33 Rab8a components (Extended Data Fig. 1). This complete display was used to document heterogeneity and to ensure that the main figures did not rely only on visually prominent examples.

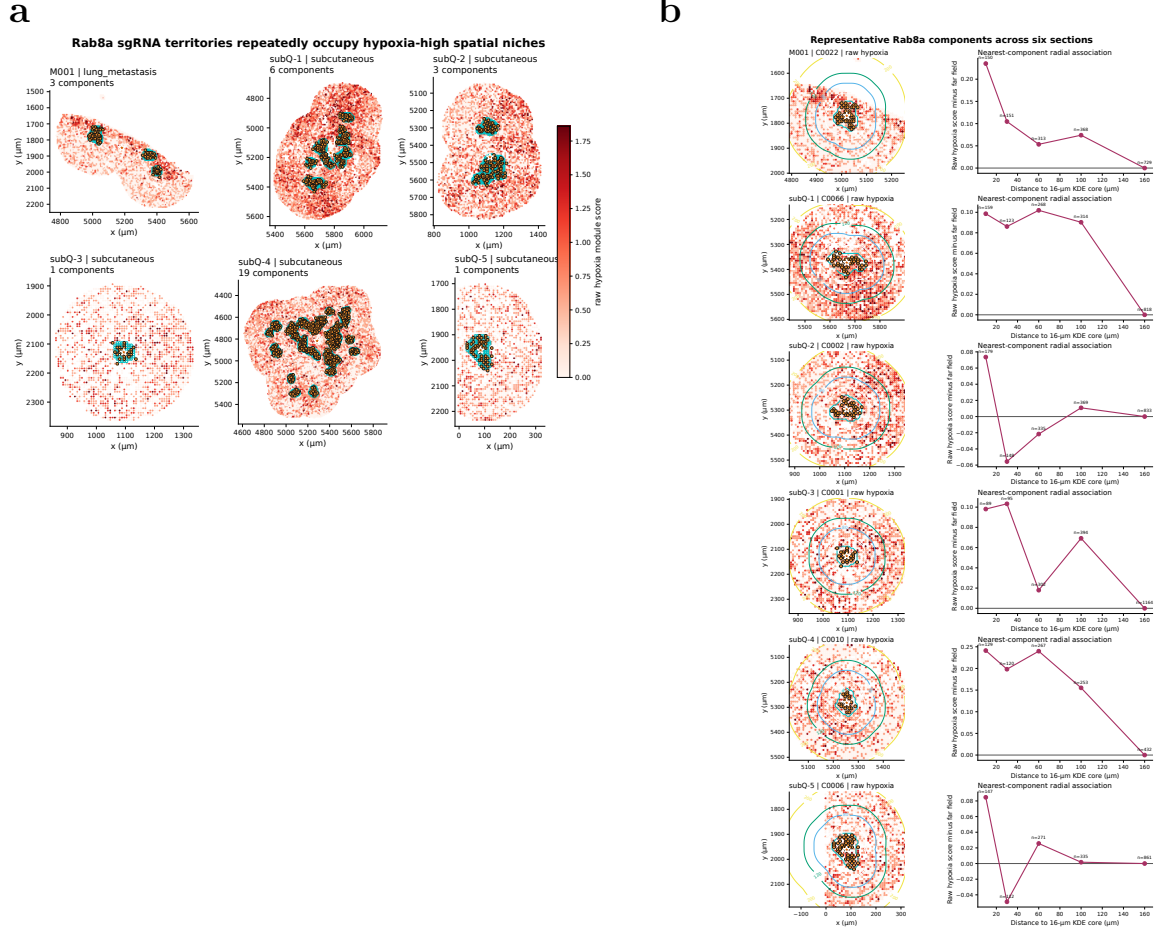


Figure 2. Rab8a source territories occupy hypoxia-high regions across tumour sections. **a**, Cross-section spatial overview for one lung-metastasis section (M001) and five subcutaneous tumour sections (subQ-1–subQ-5). Each background point is an analysed spatial bin coloured by its unsmoothed raw hypoxia-module score. The colour map is fixed to Reds with a lower limit of zero and a shared upper limit at the 99th percentile of all displayed hypoxia scores, reducing the influence of extreme values while retaining a common scale across sections. Orange points mark observed sgRab8a_1-positive malignant source bins, cyan contours denote the core of the fixed 16- μ m source-only KDE field, and red stars indicate component centroids. Source markers are plotted over the expression layer so that their coordinates remain visible. **b**, Representative Rab8a components selected by spatial support rather than response magnitude. The left column shows local tissue coordinates, measured source bins, KDE cores and distance contours; the right column shows the component-specific mean raw hypoxia score in the five pre-specified distance rings after subtraction of the component’s 120–200- μ m far-field mean. Numbers beside radial points indicate available target-bin support. Expression was not smoothed, and distance was measured from the KDE-defined source core rather than from the centroid.

2.3 The structured spatial null limits a causal interpretation

The observed positive association alone cannot distinguish Rab8a-dependent biology from preferential source localization in a structured tumour niche. We therefore permuted intact component labels within section while retaining component positions and approximate geometry. The spatial-null distribution was shifted in the same direction as the observed association: its median was 0.0133, compared with an observed coefficient of 0.0175. The Rab8a-specific excess over the null median was 0.0042, with an empirical permutation P value of 0.170 and a false-discovery rate of 0.647.

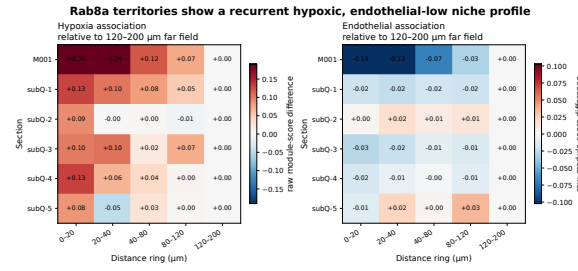
Radial summaries also varied among components (Fig. 3a). Some source components showed a near-field hypoxia enrichment that weakened with distance, whereas others were non-monotonic. The endothelial response showed an overall negative direction but similarly heterogeneous local geometry. These features support interpretation as a recurrent spatial niche association rather than a uniform isotropic propagation kernel.

Section-level joint coefficients reinforced the directional contrast while revealing differences in amplitude (Fig. 3b). M001 showed the largest positive hypoxia coefficient, whereas several subcutaneous sections showed smaller positive values. Endothelial coefficients were negative in every section, but the gene-specific deviation from the endothelial spatial null was small (-0.00137 ; permutation $P = 0.555$). Thus, consistency of direction was stronger than evidence for Rab8a specificity. We interpret the endothelial result as part of the descriptive niche state and not as an independently significant downstream response.

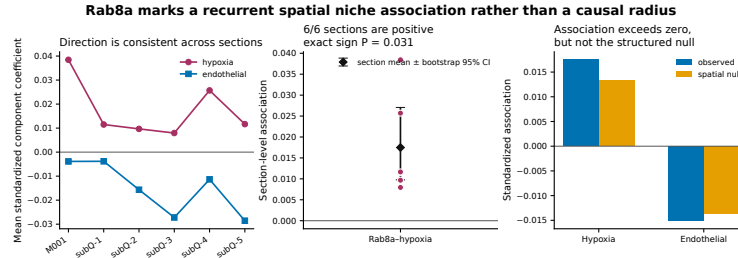
The distance-by-angle analysis provided a direct diagnostic of isotropy (Fig. 3c). Hypoxia enrichment was strongest in selected near-field angular sectors and attenuated un-

140 evenly across direction. A spatially propagating diffusible response could be anisotropic
141 because of tissue barriers or cellular relays, but anisotropy can also arise from the shape of
142 a pre-existing niche or incomplete tissue support. In the absence of independent guides and
143 matched controls, the angular pattern therefore strengthens the niche interpretation rather
144 than rescuing a radius estimate.

a



b



c

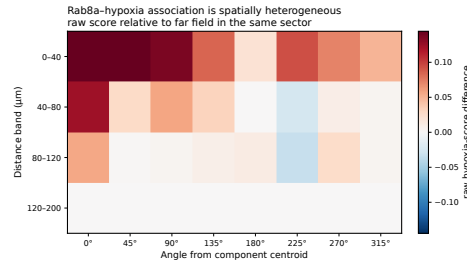


Figure 3. Rab8a is associated with a spatially heterogeneous hypoxia-high, endothelial-low niche. **a**, Section-by-distance matrices for hypoxia (left) and endothelial (right) response programmes. Rows correspond to the six analysed sections and columns to the five distance rings. Each cell is the section-level mean unsmoothed programme score after subtraction of the corresponding 120–200- μm far-field mean; the far-field column is therefore zero by construction. Red values indicate enrichment relative to the far field and blue values indicate depletion. **b**, Statistical summary of the Rab8a association. The left panel shows section-level joint coefficients for hypoxia and endothelial programmes, retaining each section as one displayed unit. The middle panel summarizes the Rab8a–hypoxia coefficient and its descriptive section-level uncertainty. The right panel compares the observed gene-level standardized coefficient with the median of the structured component-label permutation null. For hypoxia, the observed effect was 0.0175, the null median was 0.0133, the gene-specific deviation was 0.00415, the empirical permutation P value was 0.170 and the replicated false-discovery rate was 0.647. For the endothelial programme, the observed effect was -0.0151 , the null median was -0.0137 and the permutation P value was 0.555. Permutations retained section and approximate component structure. **c**, Distance-by-direction matrix for raw hypoxia association. Columns divide the neighbourhood into eight 45° angular sectors and rows show radial distance bands. Values are raw hypoxia-score differences relative to the far field in the same sector. The directional variation demonstrates that the local state is not spatially uniform; it is used as a geometric diagnostic and not as an estimate of causal propagation.

3 Discussion

Our analysis identifies a recurrent Rab8a-associated tumour state. Across six sections and two tissue contexts, Rab8a-labelled malignant territories occurred in neighbourhoods with higher hypoxia scores and lower endothelial scores. Hypoxia is a major ecological pressure in solid tumours and is closely linked to vascular organization, tumour evolution and treatment response[10, 11]. The spatial phenotype observed here was visible in tissue maps and directionally consistent at the section level, making Rab8a a reasonable candidate for prospective validation.

The data do not demonstrate that Rab8a perturbation generated this state. First, the structured spatial null was already shifted towards a positive hypoxia association. Second, only one Rab8a sgRNA was represented. Third, matched non-targeting-control territories and animal identities were unavailable. Fourth, component-level profiles were spatially heterogeneous and not uniformly monotonic. Preferential survival or expansion of Rab8a-labelled cells in a pre-existing hypoxic niche therefore remains a viable explanation.

These limitations also determine what should not be reported. The 120–200- μm reference ring is a statistical comparator, not an estimated radius. An exponential length scale, R_{50} or $R_{5\%}$ would imply a causal decay process that the present design cannot identify. More complex machine-learning models cannot recover missing experimental contrasts and may amplify leakage if neighbouring bins from the same source territory are divided between training and test sets.

A decisive validation experiment should include at least two independent Rab8a sgRNAs, multiple non-targeting controls, independent animals and orthogonal measurements

of hypoxia and vascular state. Rescue or inducible perturbation would help distinguish niche selection from perturbation-induced change. Analysis should hold out entire animals or sections and test whether the Rab8a distance-response curve exceeds a spatially structured control distribution. Only after these conditions are met would fitting a biological propagation kernel be justified.

4 Methods

4.1 Data and study design

This study is a computational reanalysis of SPAC-seq tumour data distributed through the SPAC data portal (<https://spac.pku-genomics.org/>) and described in the primary SPAC-seq study[7]. No new animal or human experiment was performed. Six sections with matched guide, transcriptome and spatial-coordinate outputs were included: M001 from the lung-metastasis cohort and subQ-1–subQ-5 from the subcutaneous cohort. The official coordinate metadata indicated an 8- μ m bin grid. Coordinates were converted to micrometres before component calling and distance calculation. The analysis used fixed random seed 20260803.

4.2 RNA processing and response scores

Raw RNA counts were normalized within bin to a common library-size total and transformed with $\log(1 + x)$. Working cell-state and response scores were calculated using fixed marker-gene modules defined before the Rab8a visualization step. The primary response was a

186 malignant hypoxia programme; the endothelial programme was retained as a complementary
187 tissue-state readout. Module scores were arithmetic means of available normalized marker
188 expression. No spatial smoothing, image-derived interpolation or neighbour averaging was
189 applied to target expression or module scores in the primary analysis. Bins lacking sufficient
190 response support remained missing rather than being assigned a value of zero.

191 **4.3 Context encoding and matched controls**

192 The general model represents the local background of each analysis bin with an encoded
193 context vector. Expression and cell-type neighbourhoods are processed by separate pool-
194 ing operations and multilayer perceptrons, and the resulting representations are combined
195 to capture molecular and compositional context without directly using the response being
196 tested. Eligible controls are selected from the same tissue section, cell type and niche. To re-
197 duce contamination by the focal source field, candidates within 400 μm of the analysis source
198 are excluded; candidates carrying more than one guide assignment are also excluded. The
199 ten nearest eligible controls in context space are retained. Matching is used to improve com-
200 parability of the observed tissue background, but does not by itself establish exchangeability
201 or biological replication.

4.4 Additive spatial exposure and regression

Source clones are represented by their spatial centroids and reconstructed source support. For analysis bin i , exposure to multiple clones is summarized as

$$E_i = \sum_c M_c K(d_{ic}; \lambda), \quad (1)$$

where M_c denotes the contribution of source clone c , d_{ic} is the physical distance between bin i and clone c , and $K(\cdot; \lambda)$ is a distance-decay kernel with scale parameter λ . The general exposure-response analysis considers bins within $300 \mu\text{m}$ of a source. Source membership T_i , spatial exposure E_i and encoded context z_i enter the regression

$$y_i = \alpha + \beta T_i + \delta E_i + \gamma^\top z_i + \epsilon_i. \quad (2)$$

Here β describes the association within source bins and δ describes the association between source exposure and the response in surrounding bins after context adjustment. These coefficients are interpreted as associations unless the guide, control and biological-replication requirements for causal identification are satisfied. The Rab8a visual and radial analysis reported below uses a conservative $200\text{-}\mu\text{m}$ window and structured spatial-label null to avoid assigning a causal radius when those requirements are unmet.

4.5 Guide assignment and source definition

For each bin, total guide UMI count, the guide with the largest count and the fraction of guide UMIs assigned to that top guide were calculated. A confident guide assignment

required at least two guide UMIs and a top-guide fraction of at least 0.70. Failure to detect a guide was not interpreted as proof that a malignant bin was unperturbed. Candidate source bins were restricted to the malignant-like compartment and to Rab8a as the mapped guide target. Spatial source components were called separately within each section and target gene using density-based spatial clustering (DBSCAN)[12] with $\epsilon = 24 \mu\text{m}$, a minimum of eight local source bins and a minimum retained component size of 15 bins. Calling targets separately prevented source components carrying different perturbation labels from being merged. Thirty-three Rab8a components passed these working criteria.

4.6 Source-field reconstruction

Observed guide-positive coordinates were converted into a continuous source-density field using an isotropic Gaussian kernel[13] with bandwidth $16 \mu\text{m}$, corresponding to two spatial bins. The primary source core was defined at a density fraction of one third of the component maximum. This translation follows the source-smoothing principle used for spatially intermittent sgRNA detection while keeping the bandwidth explicit in physical units. The kernel was applied only to the perturbation source field. Target expression values and pathway scores were not smoothed. Source coordinates, KDE-core membership and centroid coordinates were saved separately so that the measured and reconstructed components could be distinguished in every figure.

4.7 Distance assignment and response programmes

All eligible non-source bins within 200 μm of a Rab8a source field were queried against source-core coordinates. Each bin was assigned to the nearest Rab8a component in the same section, producing one primary owner and avoiding repeated counting across overlapping Rab8a neighbourhoods. Minimum Euclidean distance to the source core was grouped into 0–20, 20–40, 40–80, 80–120 and 120–200- μm rings, with midpoints 10, 30, 60, 100 and 160 μm used only for plotting. For each component and ring, the analysis recorded the number of eligible target bins and the mean unsmoothed response score. Component-relative profiles were calculated by subtracting the component-specific 120–200- μm mean. Section-level radial profiles pooled eligible target bins within section before the same subtraction. The outer ring was a statistical reference and was not considered a measured radius.

4.8 Joint spatial association model

For each response, component-level spatial associations were estimated jointly with section context so that differences in baseline programme score among tissue sections were not interpreted as Rab8a effects. Gene-level standardized effects were aggregated from evaluable components, and the direction of the section-specific coefficients was retained as a replication diagnostic. Six sections and 33 components were evaluable for the primary Rab8a–hypoxia analysis. Because all components carried sgRab8a_1, guide nested within target gene could not be estimated.

4.9 Statistical analysis

Descriptive uncertainty was evaluated by resampling sections and then components within resampled sections. The section bootstrap summarizes stability across the available tissue sections but was not interpreted as animal-level inference because section-to-animal identities were unavailable. For the structured null, intact source components retained their position, size and shape while perturbation labels were exchanged within section and approximate component-size strata. The primary analysis used 9,999 permutations. The two-sided empirical P value measured the position of the observed gene-level coefficient relative to this structured null. Multiple testing across the evaluated gene-response family was controlled with the Benjamini-Hochberg procedure[14]. Bins were never treated as independent biological replicates.

For the angular analysis, each target bin was assigned to one of eight 45° sectors around its nearest component centroid and to a radial distance band. Sector-specific raw hypoxia means were centred against the far field from the same angular sector. This analysis was used to detect directional heterogeneity and incomplete isotropy. No exponential, Hill or spline-derived radius was reported because the observed association did not exceed the structured null after multiple-testing correction, only one Rab8a guide was available and the component curves were not uniformly monotonic.

4.10 Visualization and reproducibility

Spatial hypoxia maps used the Reds colour map with $v_{\min} = 0$ and a common $v_{\max}$ equal to the 99th percentile of all plotted hypoxia scores. Observed source bins, KDE-core contours

and component centroids were overlaid as separate layers. The complete set of 33 components was exported in addition to representative examples to reduce visual selection bias. Panel-level source data were written as tab-separated tables, including component radial profiles, section radial profiles, joint coefficients, permutation summaries and distance-by-angle values.

5 Data availability

The SPAC-seq datasets analysed in this study are available at <https://spac.pku-genomics.org/>.

6 Code availability

Analysis code and tutorials are available at https://github.com/Xu-zhenghao-zzz/2026_pebble/tree/v2-perturbgmn-20260802.

7 References

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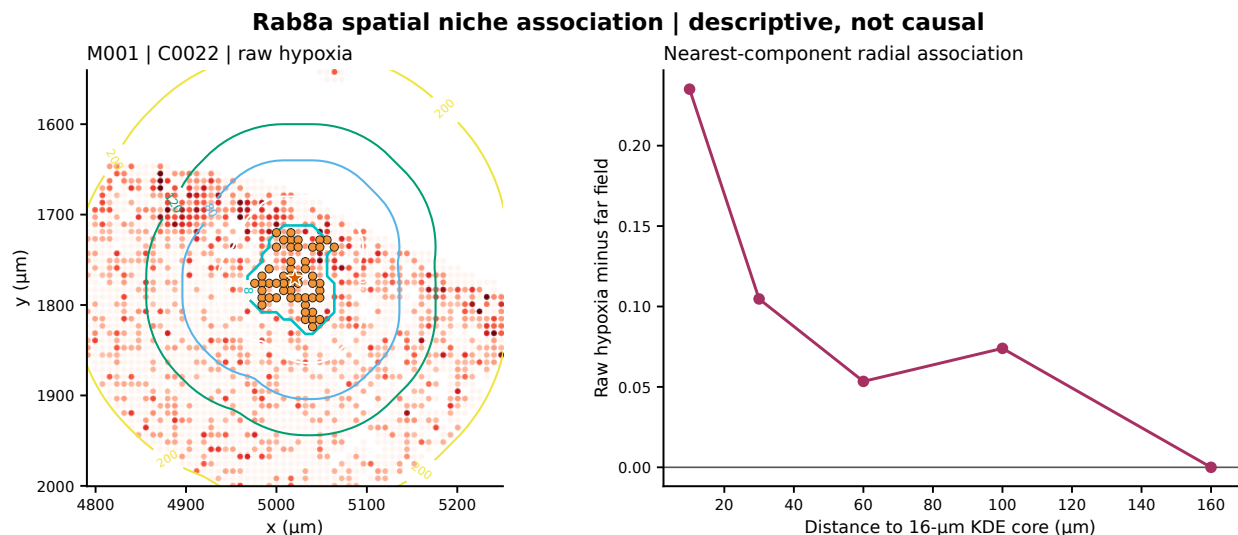
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Extended Data Figure 1



Extended Data Fig. 1. Complete spatial and radial audit of all Rab8a source components. Each page corresponds to one of the 33 Rab8a components retained across M001 and subQ-1–subQ-5. The left panel shows the local tissue-coordinate map with unsmoothed raw hypoxia scores, observed sgRab8a_1-positive malignant source bins, the 16-μm source-only KDE core and the component centroid. The right panel shows mean raw hypoxia score in the 0–20, 20–40, 40–80, 80–120 and 120–200-μm rings after subtraction of the component-specific far-field mean; numbers identify available target-bin support. Components are included irrespective of effect direction or visual prominence. Variation in curve shape, target support and local tissue geometry documents the heterogeneity that prevents a single isotropic Rab8a propagation radius from being assigned.